Paleomagnetism of Lower and Middle Cambrian Sedimentary Rocks From the Desert Range, Nevada

STEPHEN L. GILLETT

Department of Earth and Space Sciences, State University of New York at Stony Brook, Stony Brook, New York 11794

DAVID R. VAN ALSTINE

Division of Geological and Planetary Sciences, California Institute of Technology, Pasadena, California 91125

Over 500 oriented samples were collected from a homoclinal sequence of Lower and Middle Cambrian terrigenous and carbonate miogeoclinal strata in the Desert Range, Nevada. Extensive thermal demagnetization was successful in isolating similar characteristic magnetization directions in red-purple mudstones of the Wood Canyon Formation (Lower(?) and Lower Cambrian) and in gray limestones of the Carrara and Bonanza King Formations (Lower and Middle Cambrian). Lithologic and magnetic evidence suggest that these magnetizations were acquired penecontemporaneously with deposition. The similarity of the characteristic magnetization directions in these strata implies that little apparent polar wander occurred with respect to North America from early Early through middle Middle Cambrian time. The divergence of these directions from those from the partly coeval Tapeats Sandstone of the Colorado Plateau probably resulted from a net 36° clockwise rotation of the Desert Range section about a vertical axis. This rotation is probably due to mid-Tertiary oroflexural bending but may in part have been caused by Mesozoic thrusting. The polarity of the geomagnetic field in early Middle Cambrian time appears to have been continuously reversed. Stratigraphic intervals of mixed polarity, possibly correlative with those found in the Tapeats, may be present in the Lower Cambrian Zabriskie Quartzite.

INTRODUCTION

A fairly extensive body of paleomagnetic data from rocks of latest Precambrian and earliest Paleozoic age from the Gondwana continents [*McElhinny and Embleton,* 1976] suggests that rates of apparent polar wander (APW) were much higher during the late Precambrian and Early Paleozoic than in the Mesozoic and Cenozoic. On this basis, *McElhinny et al.* [1974] predicted that 'extensive studies of late Precambrian and Cambrian rocks of North America . . . will reveal a complicated apparent polar wander path with a polar shift of at least 180°.' Whereas a relatively continuous APW path can be drawn for North America since the Early Ordovician [*Van Alstine and de Boer,* 1978], the APW path of the latest Precambrian and Cambrian remains poorly known.

To help extend the continuous North American APW path into these times, we have undertaken a paleomagnetic investigation of upper Precambrian and Lower Paleozoic strata in the Cordilleran Geosyncline. A miogeoclinal sequence exposed in the Desert Range, Nevada, was chosen for study because it is conformable from the upper Precambrian through Lower Ordovician, occurs in a generally homoclinal fault block, shows little if any evidence of regional metamorphism, and is well exposed. Besides the advantage of stratigraphic continuity, this miogeoclinal sequence is known to include the Precambrian-Cambrian boundary, unlike any sequence on the craton. A disadvantage of this section is that it lies within a structurally complex area, so that the probability of tectonic rotations about a vertical axis must be reckoned with. Such rotations may be corrected for, however, by calculating the declination correction required to make paleomagnetic poles from these rocks match poles of the same age from the craton.

The work reported in this and the companion paper [*Van Alstine and Gillett,* 1979] represents results from over 800 independently oriented samples collected in the Desert Range during four sampling trips over 2 years. Of this total, about one third come from upper Precambrian strata, and two thirds from Lower and Middle Cambrian strata. This paper focuses on the paleomagnetism of the Cambrian part of the section, with particular emphasis on determining whether there has been postmagnetization rotation of the fault block about a vertical axis. The companion paper includes results from the upper Precambrian part of the section. Both papers seek to elucidate the pattern of geomagnetic reversals for their possible utility in global stratigraphic correlation.

REGIONAL GEOLOGIC SETTING OF THE DESERT RANGE

The Desert Range is located in Clark County, southeastern Nevada (Figure 1). It is an eastward dipping block which is much cut by faulting and which has been interpreted by *Longwell* [1945, 1960] to be the eastern limb of a broad north plunging anticline. Immediately east of the Desert Range is the Sheep Range, part of a huge allochthonous plate which has overridden rocks of the Las Vegas Range, farther to the east, on the Gass Peak Thrust. The large-scale thrusting occurred during the Sevier Orogeny in Late Cretaceous time [*Fleck,* 1970]. *Burchfiel et al.* [1970] have shown that a belt of older (probably late Triassic to early Jurassic) thrust faults is exposed about 25 km west of the Desert Range. Normal faults, related to late Tertiary Basin and Range extension, separate the downfaulted east side of the Desert Range from the Sheep Range, and similar faults probably define the western side of the range. Because of its position with respect to the Mesozoic thrust belts, the Desert Range is probably allochthonous, but the inferred basal thrust is hidden.

To the south the Desert Range is truncated by the Las Vegas Shear Zone, a fault zone of right lateral displacement which is a major tectonic element in southern Nevada [*e.g., Longwell,* 1960, 1974; *Stewart,* 1967]. The time of the major movement on this shear zone probably lies in the range 17–11 m.y. B.P. [*Ekren et al.,* 1968; *Longwell,* 1974]. The pronounced westward bend of the strike of the beds at the southern tip of the range has been interpreted as drag on this shear zone [cf. *Burchfiel,*

Copyright © 1979 by the American Geophysical Union.

Paper number 9B0430.
0148-0227/79/009B-0430$01.00

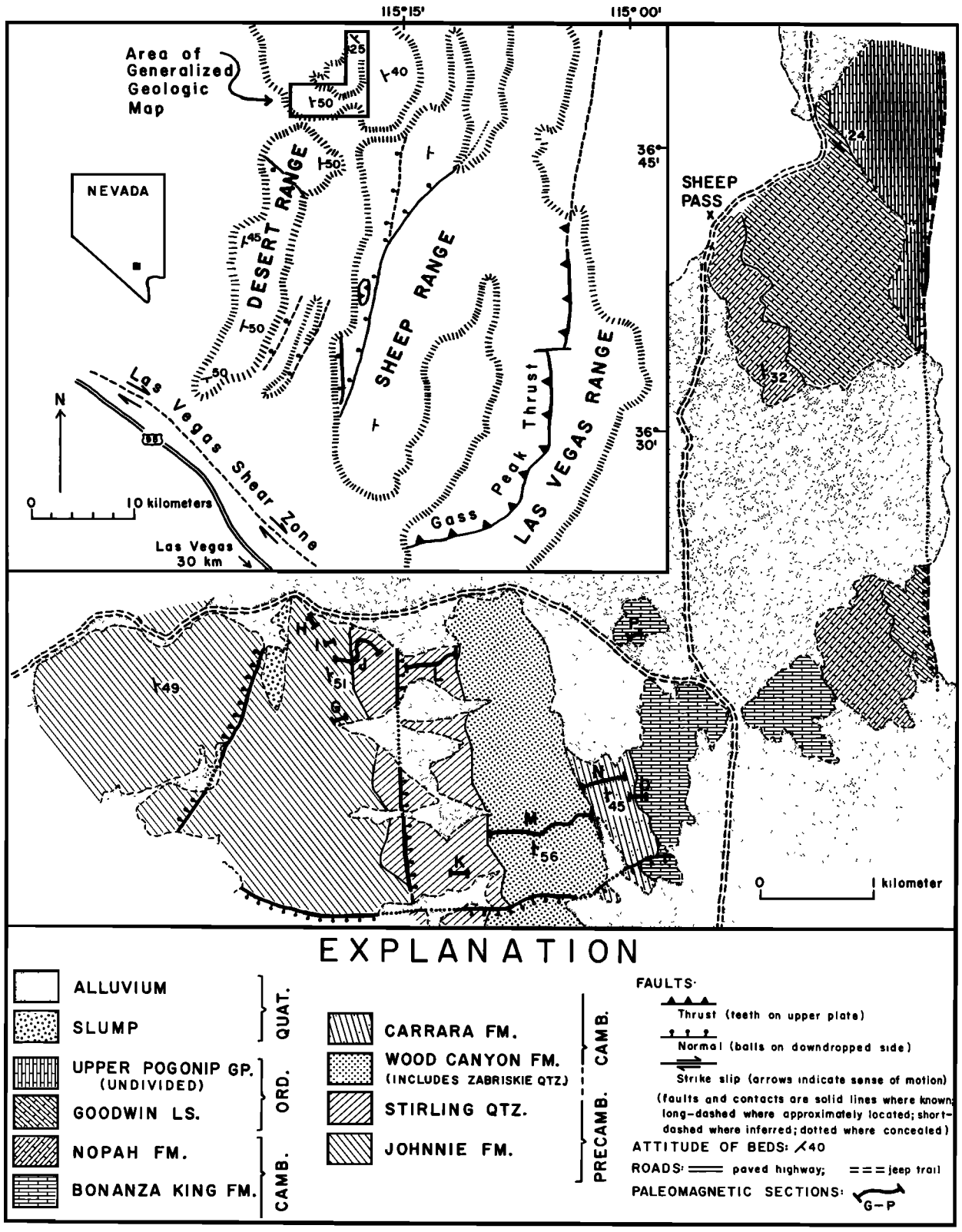


Fig. 1. Generalized tectonic and geologic maps showing the location of upper Precambrian and Cambrian paleomagnetic sections. Geologic mapping by D. R. Van Alstine, J. L. Kirschvink, and S. L. Gillett (1975–1977).

1965; *Longwell et al.*, 1965; *Albers*, 1967; *Stewart*, 1967]. These authors have also commented on similar westward bends in the structural grain and in isopach trends of strata in ranges nearby. *Albers* [1967] and *Stewart* [1967], in particular, have presented evidence for major right lateral displacement and large-scale oroflexural bending in this part of the Basin and Range Province; Albers attributes arcuate trends of many of the ranges to such oroflexural bending. It seems likely that the moderate northwesterly change in trend of the Desert Range north of the area sampled, reflecting a change in strike of the underlying strata, also represents an oroflexural bend.

Upper Precambrian and Paleozoic rocks exposed in the Desert Range are miogeoclinal deposits typical of the Cordilleran Geosyncline [*Stewart*, 1970]. Deposition in the geosyncline

21562202b, 1979, B9, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/JB084iB09p04475 by Curtin University Library, Wiley Online Library on [05/05/2025]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

was essentially continuous from late Precambrian into Early Permian time; and the total thickness of strata in this area probably exceeds 9 km [cf. *Longwell et al.*, 1965; *Burchfiel et al.*, 1974]. The section studied is a spectacularly exposed homocline which includes about 6.5 km of upper Precambrian through Ordovician strata (Figure 1). In this paper we will present paleomagnetic data on rocks from the Wood Canyon Formation (Lower(?) and Lower Cambrian), the Zabriskie Quartzite (Lower Cambrian), the Carrara Formation (Lower and Middle Cambrian), and the lowermost Bonanza King Formation (Middle Cambrian).

SAMPLING, LABORATORY, AND STATISTICAL PROCEDURES

Only oriented block samples could be taken from the Desert Range, because the area is part of the Desert National Wildlife Range, where motorized coring is forbidden. We therefore devised a block sampling technique that is reasonably fast and which works equally well with either smooth rocks such as quartzites or ones with jagged surfaces, such as solution-pitted carbonates.

Glass microscope slides, trimmed to 2.5-cm squares, were glued flat with automobile weather-stripping adhesive to the blocks to be collected. The upper edge of each slide was leveled by means of a hand-held spirit level. After the adhesive had fully set, the strike and dip of the slide were measured with a Brunton compass, and the block was collected. At this point the slide was also outlined on the surface of the rock sample, with the top indicated, so that an approximate orientation could be recovered even if the slide were accidentally broken off in transit to the laboratory. This technique for sample orientation is probably more accurate than conventional block sampling, because a flat surface with a machined straightedge is affixed to each rock and oriented independently of the character of the rock surface itself.

Blocks were collected in stratigraphic sequence. The stratigraphic intervals in the initial reconnaissance were estimated by pacing, but the sections later sampled in detail were measured with a tape and/or Jacob's staff.

All laboratory work was performed at the U.S. Geological Survey Paleomagnetics Laboratory at Flagstaff, Arizona. Generally, two cylindrical cores 2.5 cm in diameter were drilled from each block sample perpendicular to the glass slide. For measurement, only a single specimen 2.5 cm long was trimmed from a core, although duplicate specimens for additional demagnetization experiments were cut as needed. Magnetic measurements were made on an ScT superconducting rock magnetometer interfaced with a DEC PDP 8/M computer. This system has a background noise level less than 1×10^{-7} emu (1×10^{-10} A m^2) and allows immediate, on-line data reduction.

Progressive thermal and/or alternating field (af) demagnetization was performed on all samples. Thermal demagnetization was done with a noninductively wound electrical oven in which the magnetic field within the specimen area was ~20 nT. Specimens were placed in nonsystematic orientations during thermal demagnetization so that systematic spurious components could not be imparted by stray fields. Alternating field demagnetization below 50 mT (500 Oe) was performed with a Schonstedt specimen demagnetizer. To minimize acquisition of anhysteretic remanent magnetization, af demagnetization above 50 mT was carried out with a three-axis tumbler.

In this paper and its companion [*Van Alstine and Gillett*, 1979], 'geographic coordinates' and 'stratigraphic coordinates' refer to the declination and inclination of the magnetic vector in coordinate systems uncorrected and corrected for the strike and dip of bedding, respectively. 'Characteristic magnetization' is used here in the sense of *Zijderveld* [1967, p. 263] as 'that magnetization which one is apt to call primary without having definite proof.' The progressive demagnetization data from nearly all specimens were plotted by computer, using a modification of *Zijderveld*'s [1967] demagnetization diagrams similar to that employed by *Roy and Park* [1974].

Because the distributions of magnetic directions encountered in this study are skewed, application of the statistics of *Fisher* [1953] is inappropriate. This skewness results from the incomplete removal of secondary components of magnetization from all samples. The mode rather than the mean of such distributions provides a better estimate of the characteristic magnetization direction as long as the remaining remanence is dominated by the characteristic component. We have determined the mode by a computer technique described by *Van Alstine* [1979]. This technique identifies the mode of even small samples of vectors with a maximum error of 0.7°. The disadvantage of using the mode rather than the mean is that it is more difficult to estimate confidence limits for the mode. *Van Alstine* [1979] has suggested that a new statistic, designated β_{95}, can provide an estimate of the 95% confidence limits for the mode. The statistic β_{95} is derived from the largest subset of the total sample that has a mean identical with the mode of the total sample. This statistic is defined as the Fisherian half angle of the cone of 95% confidence for the mean of this subset. Implicit in this definition is the assumption that this largest subset of the total sample has a Fisherian distribution.

WOOD CANYON FORMATION

The Wood Canyon is the lowest formation in the Desert Range sequence that is demonstrably at least partly of Cambrian age on the basis of fossils it contains. The Wood Canyon is composed of a heterogeneous sequence of siltstone, quartzite, and minor carbonate rock, in which *Stewart* [1970] has described three informal members (lower, middle, and upper) of regional extent (Figure 2). At several localities where the Wood Canyon has been studied, Early Cambrian trilobites referable to *Judomia*(?) and *Nevadella* are reported from the upper half of the upper member [*Stewart*, 1970; *Palmer*, 1971]. Because the lowest stratigraphic occurrence of trilobite fossils falls within the upper member, the Wood Canyon has been assigned a latest Precambrian and earliest Cambrian age [*Stewart*, 1970]. However, if lithologic correlations of the Desert Range strata with strata in the White-Inyo Mountains of eastern California are accepted [*Stewart*, 1970; *Nelson*, 1978], the lowest trilobites (*Fallotaspis*) in the White-Inyo Mountains section occur in strata correlative with the lower and middle members of the Wood Canyon. Moreover, in the White-Inyo section, *Skolithos*, *Cruziana*, and other possibly Cambrian trace fossils occur even lower than the lowest trilobites. On this basis, *Nelson* [1978] suggested that the Precambrian-Cambrian boundary may lie in strata correlative with the upper part of the Stirling Quartzite, which conformably underlies the Wood Canyon.

Initially, about 120 samples, representing a variety of rock types, were collected from the Wood Canyon Formation at approximately 5-m intervals (section M in Figure 1). Resampling was concentrated mostly in red-purple mudstone interbeds in the middle member and in a few horizons in both the lower and upper members that had given promising preliminary results.

All the initial samples from the Wood Canyon were progressively thermally demagnetized at six steps between 300° and 610°C; most were then further demagnetized at 640°C. Some

21562202b, 1979, B9, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/JB084iB09p04475 by Curtin University Library, Wiley Online Library on [05/05/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

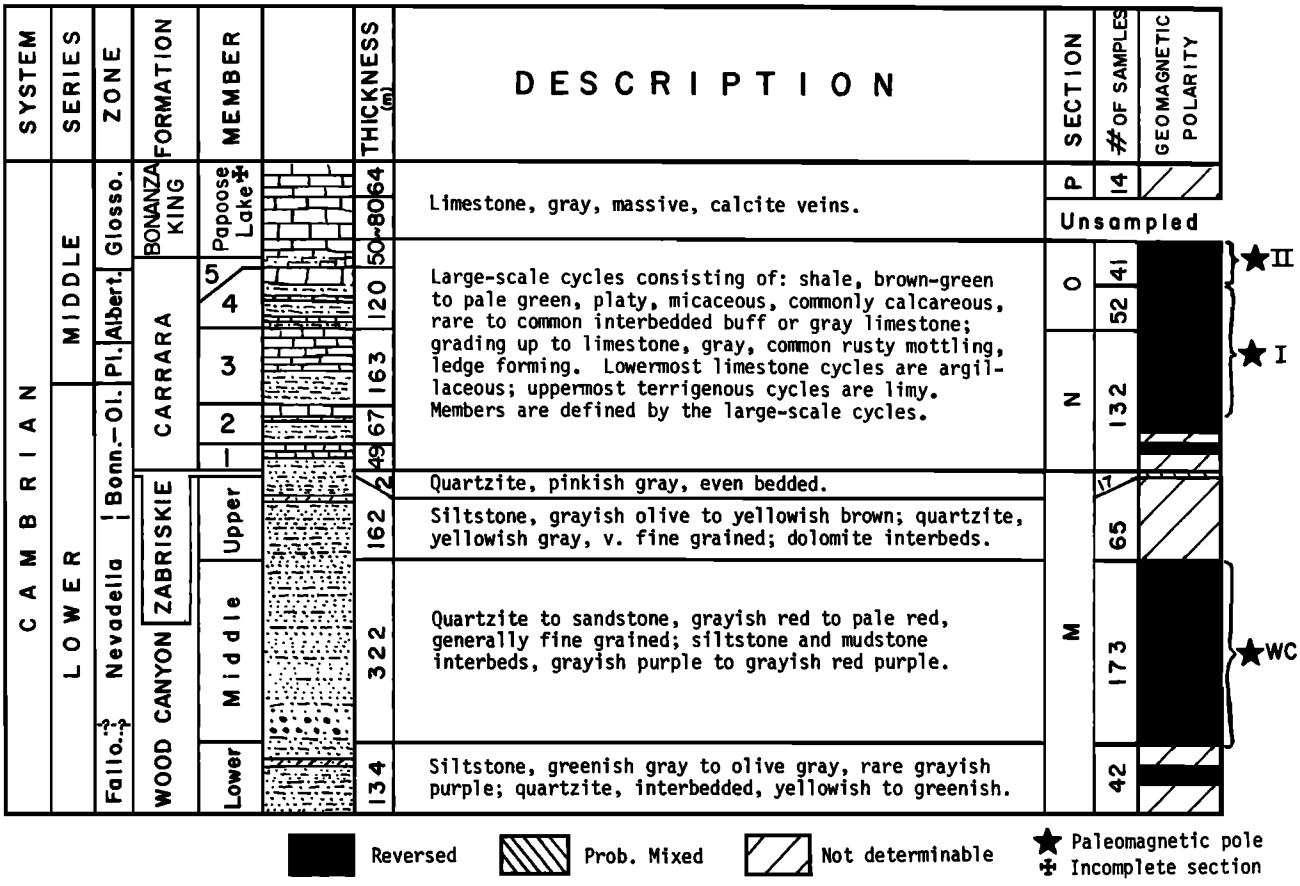


Fig. 2. Chart showing description of strata, trilobite zones, stratigraphic distribution of samples, and inferred geomagnetic polarity for Lower and Middle Cambrain formations sampled in this study. 'Fallo.,' 'Bonn.-Ol., 'Pl.' 'Albert.,' and 'Glosso.' stand for *Fallotaspis, Bonnia-Olenellus, Plagiura-Kochaspis, Albertella,* and *Glossopleura* zones, respectively. Units labeled 1–5 in the Carrara Formation are the large-scale sedimentary cycles (see text). Members composing the cycles are cycle 1, Restings Springs Member of the Zabriskie Quartzite, Eagle Mountain Shale, and Thimble Limestone; cycle 2, Echo Shale and Gold Ace Limestone; cycle 3, Pyramid Shale and Red Pass Limestone; cycle 4, Pahrump Hills Shale and Jangle Limestone; cycle 5 (incomplete), Desert Limestone. The top of cycle 5 is defined by the basal strata of the Bonanza King Formation (see text). Descriptions of Wood Canyon Formation and Zabriskie Quartzite are adapted from *Stewart and Barnes* [1966] and *Stewart* [1970]. Descriptions of the Carrara and Bonanza King formations are from S. L. Gillett and D. R. Van Alstine (unpublished data, 1976).

of the lower thermal steps were omitted for the latest samples collected, but in no case were fewer than three thermal demagnetization steps performed on any sample. In addition, 24 samples showing directions farthest from the direction of the present axial dipole field at the sampling site were further demagnetized: 12 at 650°C and the remaining 12 at 670°C. Results from the middle member, in which a characteristic magnetization could be identified, are presented first; discussion of the generally scattered directions from the rest of the formation is deferred until later.

Middle Member, Wood Canyon Formation

Natural remanent magnetization (NRM) directions for most samples from the middle member of the Wood Canyon were distributed between the present axial field direction and a direction in stratigraphic coordinates with a declination of about 110° and an inclination near 0° (Figure 3). After demagnetization at 640°C the directions had become better grouped with a mode at a low negative inclination and a declination near 125°; fewer samples showed a pronounced present axial field component. Directions from the samples further demagnetized at 650°C (not illustrated) showed increased scatter; all samples demagnetized at 670°C exhibited scattered directions and viscous components of magnetization.

In detail, the thermal demagnetization behavior of a given sample from the middle member is a function of its grain size and color. As summarized in Figure 2, the middle member of the Wood Canyon consists predominantly of pale reddish quartzites, with darker purplish and red-purple siltstone and mudstone interbeds. Most of the directions far from that of the present axial field before and after demagnetization are found in the finer-grained rocks (Figure 3). The finer-grained rocks are also the darkest (5 RP 4/2 versus 5 RP 6/2 or 5 P 6/2). In addition, samples which are red-purple (5 RP 4/2) rather than purple (5 P 4/2) tend to have directions farther from that of the present axial field.

The outliers in Figure 3 are probably caused by lightning strikes. This is suggested by the anomalously high intensities of magnetization in a number of these samples. Several of these samples were progressively af-demagnetized up to several hundred milliteslas in an attempt to remove a possible isothermal remanent magnetization, but the directions from none changed significantly with either af or thermal demagnetization. While some samples with anomalous directions do not have anomalous intensities, in general they were obtained near samples which do. A few anomalous directions are nearly antipolar to the mode defined by the mudstones, which suggests they might represent zone(s) of opposite polarity. How-

21562202b, 1979, B9, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/JB084iB09p04475 by Curtin University Library, Wiley Online Library on [05/05/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

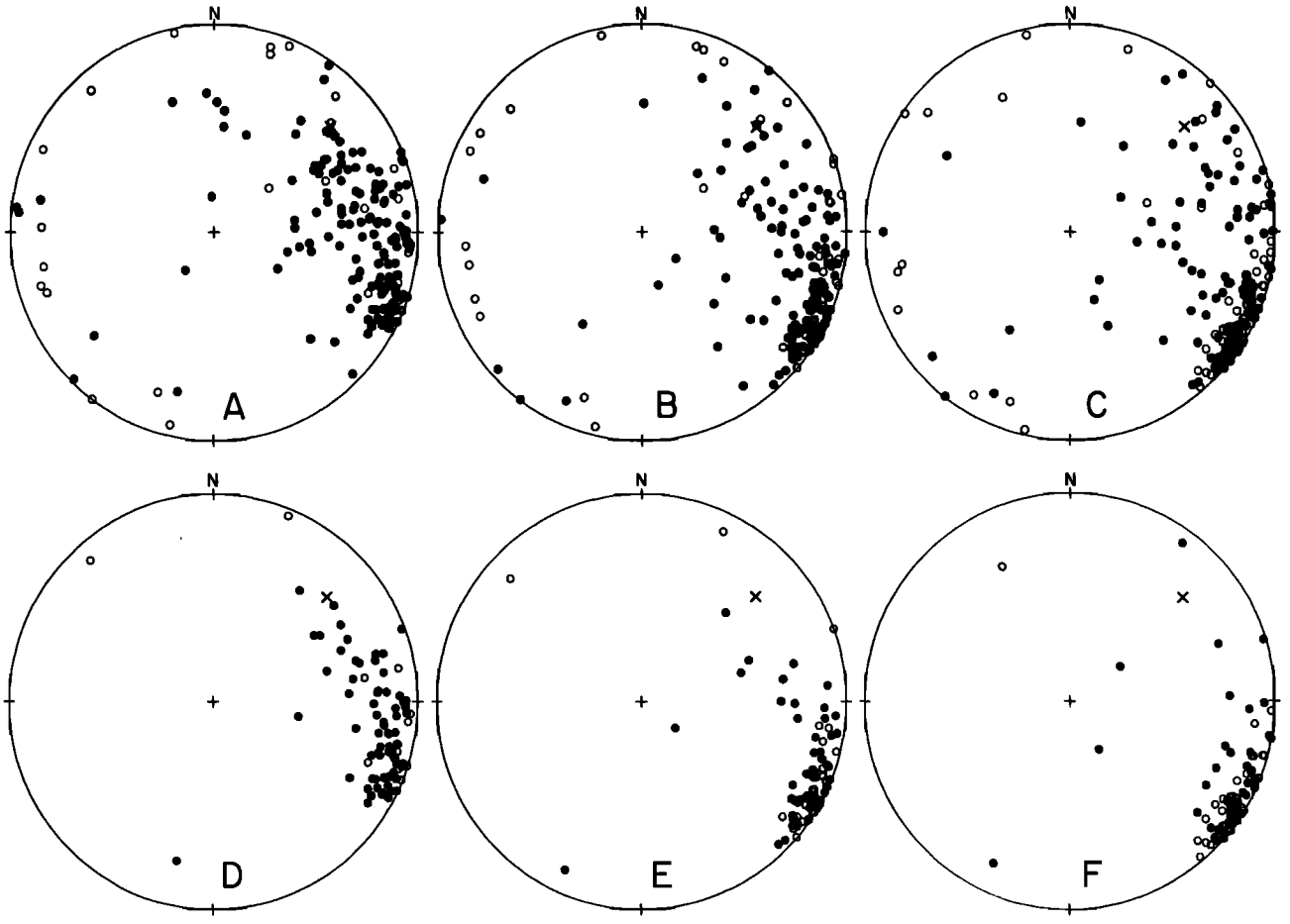


Fig. 3. Directions at selected steps from progressive thermal demagnetization of samples from the middle member of the Wood Canyon Formation. Equal-area projection in stratigraphic (dip corrected) coordinates. Open (solid) circles are on upper (lower) hemisphere, respectively. Cross marks the direction of the present axial dipole field at the sampling site after rotation for the mean attitude of beds. Directions from all samples are shown for (*a*) NRM, (*b*) 550°C, and (*c*) 640°C. Directions from red-purple mudstones only are shown for (*d*) NRM, (*e*) 550°C, and (*f*) 640°C.

ever, these directions were not reproduced in samples collected at stratigraphically equivalent horizons but off the ridge crest, where most samples of the middle member were taken.

The high blocking temperature of the characteristic magnetization of the middle member, as well as the deep red-purple color of these rocks, demonstrates that this component resides in hematite. Before this magnetization can be accepted as being representative of the Early Cambrian geomagnetic field, however, evidence for the time of its acquisition must be sought. Accordingly, we now present textural, petrographic, and stratigraphic evidence that the characteristic remanence of the middle member probably was acquired penecontemporaneously with deposition.

All samples were examined with a low-power binocular microscope. Thin sections of eight red-purple mudstones, three purplish siltstones, and one coarse-grained, pale quartzite were examined in both transmitted and reflected light.

The petrographic observations indicate that at least three types of hematite are present in the rocks of the middle member of the Wood Canyon Formation. (1) There are discrete grains of specular hematite, generally 5–25 μm in diameter. A detrital origin for these grains is inferred from rounding, which suggests transport; apparent size sorting; lack of alteration halos; occurrence of imbricate trains suggestive of heavy mineral lenses; and elongation subparallel to the bedding fabric, suggesting depositional control. (2) Another type is aggregates containing minute grains ($\sim$1 μm) of specular hematite and/or red, earthy, microcrystalline hematite, presumably formed by alteration of ferroan precursors. An authigenic origin is suggested because such aggregates apparently could not survive transport and because commonly they have diffuse contacts suggestive of alteration halos. Both of the preceding forms of hematite occur in all the rock types. (3) There is also microcrystalline red cement, occurring primarily in the red-purple mudstones. This cement, which is responsible for the deep red-purple (5 RP 4/2) color of these rocks, is largely confined to very fine-grained parts of the rock; coarser, quartzofeldspathic laminae and lenses are light colored.

All three forms of hematite may contribute to the characteristic magnetization of these rocks. The presence of detrital specular hematite in all rock types of the middle member suggests that part of the characteristic remanence may be a detrital remanent magnetization (DRM). A chemical remanent magnetization (CRM) in the authigenic specular hematite may also contribute to the characteristic magnetization; if so, constraints similar to those discussed below apply to the timing of the acquisition of this CRM. A component of the characteristic magnetization can be discerned in most specimens from the middle member, even in the pale quartzites which contain very little of the red cement. Thus the detrital specular hematite and/or the hematitic aggregates apparently contribute in part to the characteristic magnetization.

21562202b, 1979, B9, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/JB084iB09p04475 by Curtin University Library, Wiley Online Library on [05/05/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

The red cement, however, is probably the most important carrier of the characteristic magnetization. This cement is the most distinguishing feature of the red-purple mudstones and must postdate their deposition to some degree; as elaborated above, the red-purple mudstones also show the best grouping of directions, farthest from the direction of the present axial field. Together, these observations indicate that a large part of the characteristic magnetization may be a CRM residing in the red cement.

Textural relations in these mudstones support an Early Cambrian time of hematization and cementation, suggesting that any CRM associated with these processes is also Early Cambrian. Within the red-purple mudstones, bioturbation is suggested by light-colored, commonly rounded pods that are filled with coarser, quartzofeldspathic material; the bedding fabric of the mudstone bends around the pods. The pods probably represent burrows infilled with some of the overlying, coarser-grained sediment. Burrowing is also suggested by distorted, light-colored laminae that contain disrupted and bent pieces of the surrounding red-purple mudstone. Even where bioturbation is present, the contact of coarser, light-colored material and fine red-purple mudstone is sharp at the scale of the thin sections. Hence at least the precursor of the hematitic cement was present prior to the bioturbation. One would expect a pervasive, later alteration to show more diffuse contacts. Further, such an alteration should follow preferentially the coarser-grained areas, because of their higher permeability. In the middle member, however, it is the least permeable beds that are the most hematitic.

An Early Cambrian time of acquisition of the characteristic magnetization of the middle member is also likely on regional stratigraphic grounds. A superimposed, post-Early-Cambrian chemical alteration should occur somewhat sporadically and have other factors affecting its regional distribution (for example, degree of fracturing and metamorphic grade) besides the physical stratigraphy, as is indeed observed with albite metasomatism at some localities in the lower member [*Stewart*, 1970]. However, the hematitic middle member, as Stewart shows, is regionally persistent over a distance of about 400 km along the sedimentary strike and has no correlation with post-Cambrian structure. Further, the middle member is in most places overlain by several hundred meters of predominantly greenish shale, siltstone, and pale quartzite in the upper Wood Canyon, Zabriskie, and Carrara formations. These rocks, as indicated by their color, are in general more reduced than the hematite-rich middle member. They are also not very permeable; indeed, they act as barriers for groundwater movement in the present Basin and Range topographic regime [*Winograd and Thordarson*, 1968]. A post-Early-Cambrian oxidizing mechanism that would regionally penetrate these beds and oxidize the middle member, not just preferentially but exclusively, seems most implausible. Similar arguments apply to any oxidizing mechanism which might be postulated to have penetrated the strata from below. The Wood Canyon Formation lies in the middle part of a miogeoclinal sequence and was deeply buried throughout Paleozoic time; hence near-surface oxidation at any time in the Paleozoic is also implausible. Together, these observations indicate that the bulk of the hematite of the middle member is an intrinsic feature of the unit, not a product of late secondary alteration.

The characteristic magnetization direction for the middle member (declination $D = 124°$, inclination $I = -3°$; Table 1) was calculated from the mode of directions from the fine-grained rocks (the red-purple mudstones and purplish siltstones) at the 640°C demagnetization step. This mode is probably representative of the characteristic magnetization direction, since sample directions at higher demagnetization temperatures no longer show consistent changes but abruptly scatter. However, the existence of only one polarity of the characteristic magnetization removes an independent check on whether all secondary components have been removed.

The polarity of the characteristic magnetization direction of the middle member is interpreted as reversed, on the basis of its proximity to the reversed direction ($D = 93°$, $I = -4°$) from the Lower and Middle Cambrian Tapeats Sandstone of the Grand Canyon [*Elston and Bressler*, 1977]. Virtually all discrete beds of fine-grained rock thicker than 2 cm exposed at section M have been sampled, with the average sampling interval approaching 1 m at the top and base of the member, where these rock types are more abundant. Fine-grained beds are rare in the middle of the member, but the magnetization of even the coarser-grained quartzites, which were sampled at about 3-m intervals, is of reversed polarity. Thus any intervals of normal polarity that might be present in the middle member

TABLE 1. Early and Middle Cambrian Paleomagnetic Poles From the Desert Range, Nevada

Formation	Section (Figure 1)	Mean Attitude Strike	Dip	$\bar{J}_{mode}$, A m²/kg × 10^{-7}	$\bar{J}_{mode}/\bar{J}_{NRM}$	Geographic Coordinates, deg D	I	Stratigraphic Coordinates, deg D	I	β_{95} (N_{mode}/N_{tot})	Apparent North Pole	Corrected North Pole
Group II*	O	350	47°E	1.2	0.09	129	+15	130	−18	2° (47/76)	37°N, 136°E	9°N, 160°E
Group I†	N, O	351	43°E	0.4	0.05	131	+15	130	−14	4° (26/48)	36°N, 134°E	7°N, 158°E
Wood Canyon‡	M	356	58°E	40.8	0.49	139	+40	124	−3	2° (56/122)	28°N, 134°E	1°S, 157°E

$\bar{J}_{mode}$ is the geometric mean of the specific intensities of magnetization at the demagnetization step(s) upon which the mode is based (see text). Geographic coordinates are the mode of specimen directions uncorrected for the bedding attitude. Stratigraphic coordinates are the mode of specimen directions after correction for local bedding attitudes, not for the mean attitude of the entire formation. Parameter β_{95} is an estimate of the 95% confidence limits for the mode. The statistic β_{95} is derived from the largest subset (N_{mode}) of the total sample (N_{tot}) that has a mean identical with the mode of the total sample. This statistic is defined as the Fisherian half angle of the cone of 95% confidence for the mean of this subset. The apparent north paleomagnetic pole has been calculated from the stratigraphic coordinates before correction for the proposed 36° clockwise rotation with respect to the Colorado Plateau (see text). The corrected north paleomagnetic pole has been calculated from the stratigraphic coordinates after correction for 36° of clockwise rotation about a local vertical axis so that Cambrian poles from the Desert Range agree with coeval poles from the Colorado Plateau (see text).

*Directions from gray, rusty-mottled limestone of the Jangle and Desert Limestone members of the Carrara Formation, and lowermost Papoose Lake Member of the Bonanza King Formation.

†Directions from gray, rusty-mottled limestone of the Gold Ace, Red Pass, and Pahrump Hills members of the Carrara Formation.

‡Directions from red-purple mudstones and purplish siltstones of the middle member.

21562202b, 1979, B9, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/JB084iB09p04475 by Curtin University Library, Wiley Online Library on [05/05/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

of the Wood Canyon Formation would have to be thin, and probably none exist.

Lower and Upper Members, Wood Canyon Formation

The directions of magnetization of about two thirds of 42 samples taken from the lower member are much more scattered than the directions found in the middle member; in addition, intensities of magnetization are about an order of magnitude weaker. Moreover, the directions of most samples change dramatically and apparently randomly above about 600°C. Commonly, a large component of magnetization roughly aligned with the present axial field is present, particularly in the more weakly magnetized samples.

Upon thermal demagnetization, directions from the remaining third of the samples from the lower member are distributed between the characteristic magnetization direction of the middle member and the direction of the present axial field. These samples were taken from an approximately 70-m stratigraphic interval in the middle of the lower member (Figure 2). The rock types present in this interval are mostly green siltstone with purplish laminae, interbedded with light-colored to purplish quartzites. Directions of magnetization similar to the characteristic magnetization of the middle member are observed only in the purplish rocks. The characteristic magnetization direction of this interval is poorly defined, apparently owing to the presence of a significant present axial field component even at the highest temperature step (640°C). In this respect these samples resemble the quartzites from the middle member. Although the direction of characteristic magnetization of the middle interval of the lower member could not be precisely determined, the polarity of all 15 samples from this interval is unambiguously reversed.

In samples from the upper member of the Wood Canyon, as in those from much of the lower member, intensities of magnetization are weaker than for the middle member, directions are scattered and change erratically upon demagnetization, and a present axial field component predominates.

ZABRISKIE QUARTZITE

At most localities in the southern Great Basin, the contact of the Zabriskie Quartzite with the underlying Wood Canyon Formation is gradational. In the Desert Range, however, the contact may represent a depositional hiatus [*Stewart*, 1970], since it is sharp, and the Zabriskie is exceptionally thin. The Zabriskie, as Stewart recognized it in the Desert Range, is a fine-grained, pinkish gray, even-bedded quartzite about 2 m thick. *Palmer and Halley* [1979] have reassigned beds gradational in lithology between the Zabriskie and the overlying Eagle Mountain Shale Member of the Carrara Formation to the Zabriskie, naming them the Emigrant Pass Quartzite Member of the Zabriskie Quartzite. The beds assigned to the Emigrant Pass are about 10 m thick in the Desert Range [*Palmer and Halley*, 1979]. In this paper, Stewart's original definition of the Zabriskie will be used, and the Emigrant Pass Member will be discussed with the Carrara Formation.

Seventeen samples were obtained from the 2-m-thick Zabriskie; this represents a higher sampling density than for any other formation in the Desert Range. These samples were progressively thermally demagnetized at six steps between 400° and 640°C; a few were initially demagnetized at 300°C as well.

The NRM directions of about half the Zabriskie samples were near the direction of the present axial dipole field. These directions changed only slightly upon thermal demagnetization, even at the highest temperature steps. Nevertheless, vector demagnetization diagrams show that demagnetization paths for nearly all the samples do not trend toward the origin but have declinations consistently displaced toward the northwest. Hence these rocks contain some component of magnetization in addition to the component aligned roughly with the present axial field. This component is not well defined, but it could have a direction antipolar to the characteristic magnetization direction from the middle member of the Wood Canyon Formation.

In the remaining half of the Zabriskie samples, thermal demagnetization more convincingly revealed a component not aligned with the direction of the present axial field (Figure 4). The direction from one sample is consistently east to southeast with shallow inclination, near the characteristic magnetization direction from the middle member of the Wood Canyon. Directions from the other samples, stratigraphically both above and below this sample, generally trend toward a direction antipolar to the characteristic direction from the middle member. These data suggest that zones of both normal and reversed polarity may be present in the Zabriskie. Because the number of samples obtained was small, the direction of characteristic magnetization of the Zabriskie cannot be defined precisely.

Thin sections of four Zabriskie specimens were examined in both transmitted and reflected light. Specular hematite, composing about 1% of the rock, is the dominant opaque mineral which is observed as discrete grains. These grains range from 1 μm to several micrometers across. As in the middle member of the Wood Canyon Formation, many of the grains are apparently detrital and can be readily distinguished from aggregates of hematite, also occurring in these rocks, which appear to have replaced ferroan precursors. Overall, the samples from the Zabriskie resemble the quartzites from the middle member

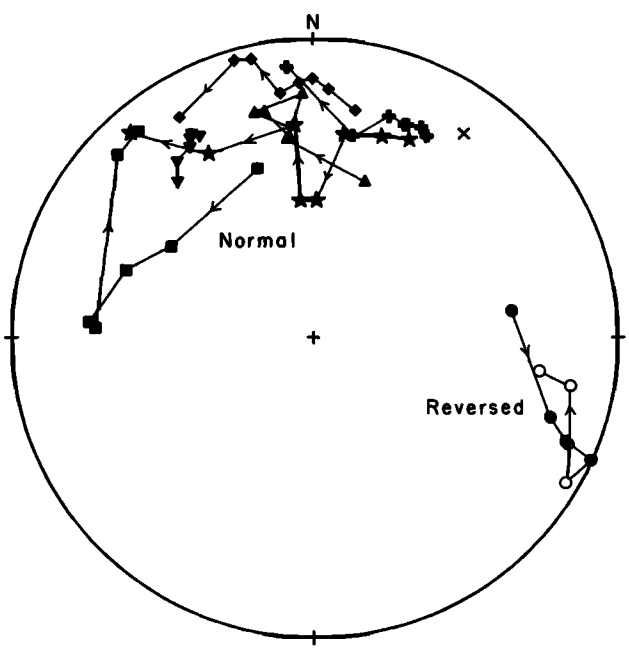


Fig. 4. Progressive thermal demagnetization behavior of selected samples from the Zabriskie Quartzite, showing one reversed and six possibly normal-polarity directions. The cross is as in Figure 3. Shown are the NRM and demagnetization steps between 300 and 640°C. Arrows point to the higher-temperature steps. Each sample is represented by a different symbol. Open (solid) symbols are on upper (lower) hemisphere, respectively.

21562202b, 1979, B9, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/JB084iB09p04475 by Curtin University Library, Wiley Online Library on [05/05/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

of the Wood Canyon Formation. Perhaps, as suggested for those rocks, the NRM includes a DRM residing in specular hematite and an early formed CRM, as well as a predominant CRM related to recent weathering.

CARRARA AND LOWERMOST BONANZA KING FORMATIONS

The Carrara Formation of Early and Middle Cambrian age conformably overlies the Zabriskie Quartzite. In turn, the Carrara is overlain with a gradational contact by the Bonanza King Formation, an immense section of carbonate rock of Middle and early Late Cambrian age. Thus deposition of the Carrara represents a transition from dominantly terrigenous sedimentation in the late Precambrian and Early Cambrian to carbonate sedimentation in later Paleozoic time.

Several sedimentary cycles, tens of meters thick, characterize the Carrara Formation [*Palmer,* 1971]. Each cycle includes fine-grained, terrigenous rocks (siltstones and shales, with minor quartzite) in the lower part which grade to limestone in the upper part. The basal terrigenous rocks of the overlying cycle rest with a sharp contact on the limestone of the underlying cycle. On the basis of these cycles, *Palmer and Halley* [1979] have divided the Carrara into nine members. The Eagle Mountain Shale and Thimble Limestone compose the lower terrigenous and upper limestone parts of the first cycle, respectively. (In this paper, data from Palmer and Halley's Emigrant Pass Member of the Zabriskie Quartzite will also be considered with the Carrara Formation.) The Echo Shale and Gold Ace Limestone define the lower and upper part of a second cycle; the Pyramid Shale and Red Pass Limestone compose a third cycle; and the Pahrump Hills Shale and Jangle Limestone make up a fourth cycle. The stratigraphically highest member is the argillaceous Desert Limestone, which forms the base of a fifth sedimentary cycle.

The Bonanza King Formation has been divided into a lower, Papoose Lake, member and an upper, Banded Mountain, member [*Palmer,* 1971]. In the Desert Range the Papoose Lake Member is about 550 m thick, and the Banded Mountain Member totals about 650 m (estimated from data of *Palmer* [1971]). Two short sections were sampled in the lower part of the Papoose Lake Member (Figure 2). The lower section immediately overlies the Desert Limestone Member of the Carrara Formation and roughly defines the top of the fifth large-scale sedimentary cycle. Seen from afar, this unit has a brownish cast which contrasts with dark gray limestone above; it is composed predominantly of gray, rusty-mottled limestone very similar to the limestones within the Carrara. The other section in the Bonanza King Formation was sampled in the dark gray limestone, about 80 m stratigraphically above the lower section.

The lower third of the Carrara Formation, up through the lowest part of the Pyramid Shale, contains latest Early Cambrian faunas [*Palmer,* 1971; *Palmer and Halley,* 1979] and has been assigned to the *Bonnia-Olenellus* Zone [*Nelson,* 1978]. Trilobites characteristic of the earliest Middle Cambrian are found in the lowest Red Pass Member, which has been assigned to the *Plagiura-Kochaspis* Zone [*Palmer and Halley,* 1979]. An *Albertella* Zone fauna is found from the top of the Red Pass Limestone to the basal part of the Jangle Limestone. Trilobites representative of the *Glossopleura* Zone occur in the Desert Limestone Member of the Carrara Formation and in the immediately overlying section in the Papoose Lake Member of the Bonanza King Formation [*Palmer,* 1971]; the higher section in the Papoose Lake Member probably also lies within the *Glossopleura* Zone [cf. *Palmer,* 1971].

Initially, the entire Carrara Formation and the lower section in the Papoose Lake Member were sampled at intervals of 5–6 m at sections N and O (Figure 1). About 75 samples were taken, representing a wide variety of rock types. The most promising paleomagnetic results came from gray, rusty-mottled limestones occurring in the limestone members of the Carrara and in the lower Papoose Lake section; hence effort was concentrated on this rock type during resampling, when another 150 samples were obtained.

Demagnetization data from the strata sampled at sections N and O will be presented in two groups. This is convenient because (1) the demagnetization steps of the two groups are somewhat different, (2) the magnetic behavior of the two groups, while similar, differs slightly, and (3) paleomagnetic poles are calculated differently from the two data sets. Group I comprises all the Carrara Formation up through the Pahrump Hills Shale. Group II comprises the Jangle and Desert Limestone members of the Carrara Formation and the lower section in the Papoose Lake Member of the Bonanza King Formation. All of the initial samples in group I were progressively thermally demagnetized at 200°, 300°, 400°, 500°, and 550°C; a few with sufficient remaining intensity were further demagnetized at 590° and 630°C. The group I resamples were also demagnetized at these steps, but with two additional steps at 350 and 450°C. The group II samples were all demagnetized at six or seven steps between 200° and 500°C; the initial samples had steps performed at 120° and 550°C as well.

To complement the thermal demagnetization experiments, 13 duplicate specimens, all from group I, were progressively af-demagnetized in eight steps at peak fields between 5 and 150 mT. For all rock types, af demagnetization generally resulted in a loss of intensity ranging from as little as 20% to over 90%, with little change from an NRM direction near that of the present axial field. Gray, rusty-mottled limestones and buff carbonates showed the least response to af demagnetization; they lost 25–40% of their remanence in the first few tens of milliteslas and little thereafter. Alternating field demagnetization alone provided no information regarding any characteristic magnetization that might be present in these rocks.

In contrast, progressive thermal demagnetization revealed the presence of several components of magnetization in the group I samples (Figure 5). The NRM directions were clustered near the present axial field direction. At 200°C most directions steepened slightly in inclination and moved slightly counterclockwise; in addition, their distribution became somewhat more scattered. At 400°C the directions of many samples moved toward a low negative inclination and a declination near 130°. Finally, by 550°C the component of magnetization with low inclination and declination near 130° was almost entirely removed.

In detail, the behavior of a given sample upon thermal demagnetization was a function of its lithology. Directions of magnetization from gray, rusty-mottled limestones of the Thimble, Gold Ace, Red Pass, and interbeds in the Pahrump Hills members most consistently showed the demagnetization behavior outlined above: i.e., a shift from NRM directions near that of the present axial field toward a direction with low inclination and southeasterly declination by 400°C (Figure 5). Directions from some of the quartzite interbeds in the Eagle Mountain and Echo shales are distributed like those from the gray, rusty-mottled limestones; however, a present axial field component was more dominant in the quartzites at all demagnetization steps. Directions from green and brown siltstones of the Carrara generally changed erratically upon de-

21562202b, 1979, B9, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/JB084iB09p04475 by Curtin University Library, Wiley Online Library on [05/05/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

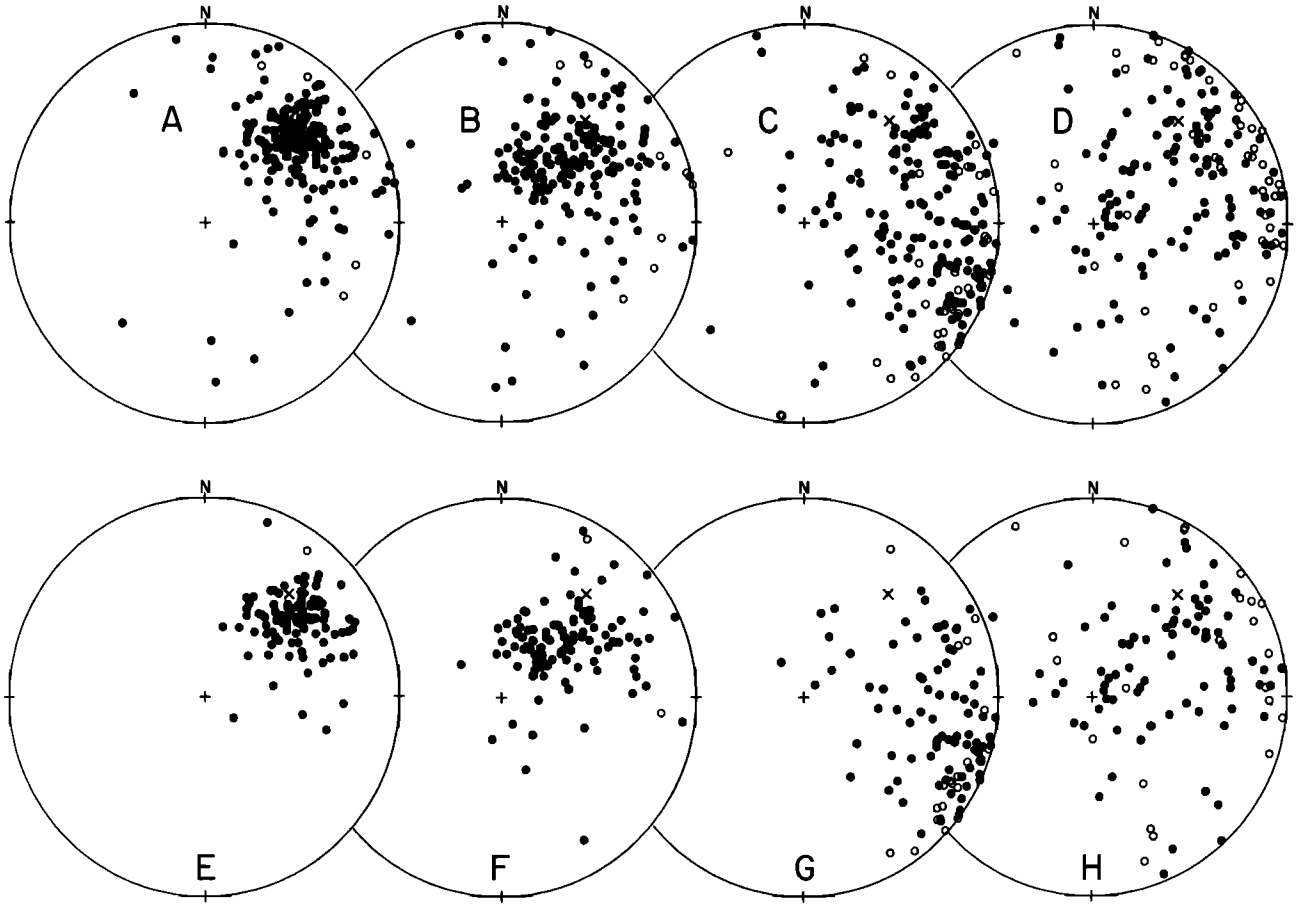


Fig. 5. Directions at selected steps from progressive thermal demagnetization of samples from group I (Carrara Formation up through Pahrump Hills Shale Member). Symbols are as in Figure 3. Directions from all samples are shown for (*a*) NRM, (*b*) 200°C, (*c*) 400°C, and (*d*) 550°C. Directions from gray, rusty-mottled limestones only are shown for (*e*) NRM, (*f*) 200°C, (*g*) 400°C, and (*h*) 550°C.

magnetization. Carbonate rocks that are buff or orange rather than predominantly gray showed present field directions which did not change appreciably upon demagnetization.

A vector demagnetization diagram of a specimen of gray, rusty-mottled limestone from group I emphasizes several distinctive features of the demagnetization behavior of this rock type (Figure 6*a*). First, about 80% of the NRM intensity is lost by 200°C, followed by a smaller loss to 400°C. Second, the direction of magnetization at 200°C is not on the great circle path from NRM to 400°C but is displaced toward a steeper inclination. Third, the demagnetization path between 400° and 500°C is essentially linear to the origin, indicating removal of a single component with moderately high blocking temperature. This demonstrates that at least three components of magnetization are present in these limestones: (1) a component which is roughly aligned with the direction of the present axial field and which has a low blocking temperature or chemical stability limit, (2) a component of intermediate thermal stability (present at ~200°C but removed thereafter), contributing to a direction with a steep inclination, and (3) a characteristic component, isolated between 400° and 500°C, with 5–10% of the NRM intensity, a stratigraphic declination near 130°, and a shallow inclination. This last component is removed by 550°C.

The group II samples showed magnetic behavior very similar to that of the gray, rusty-mottled limestones from group I (Figure 7); the three components of magnetization are again present. (A few samples, taken from a 10-m-thick whitish limestone unit in the Jangle Member, had essentially no magnetization remaining above 250°C; they are excluded from discussion and from Figure 7.) The rock types in group II are also mostly gray, rusty-mottled limestones, so the similarity in behavior is not surprising; however, the characteristic magnetization was much more effectively isolated by thermal demagnetization in the group II samples, as is shown by their better grouping at 400° and 450°C. This may be due to less deep weathering in the group II samples, leading to less remagnetization. In general, the low-temperature, present axial field component in the group II samples is much smaller, as shown by a vector demagnetization diagram of a typical group II sample (Figure 6*b*); this lends some support to this idea. However, no difference in degree of weathering was apparent in the field between the group I and group II samples.

The component in both groups I and II stable only at low temperature probably in part represents a CRM developed during recent weathering. This component may reside either in goethite (which has a Néel point near 120°C; e.g., *Strangway* [1970, p. 35]) or perhaps in very fine grained hematite; it is probably related to the rusty seams and mottling noted in hand specimen. The initial drop in intensity of magnetization upon af demagnetization (in the group I duplicate specimens), however, suggests that part of this component may be a viscous magnetization residing in magnetite.

The cluster of directions observed at 200°–250°C in groups I

21562202b, 1979, B9, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/JB084iB09p04475 by Curtin University Library, Wiley Online Library on [05/05/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

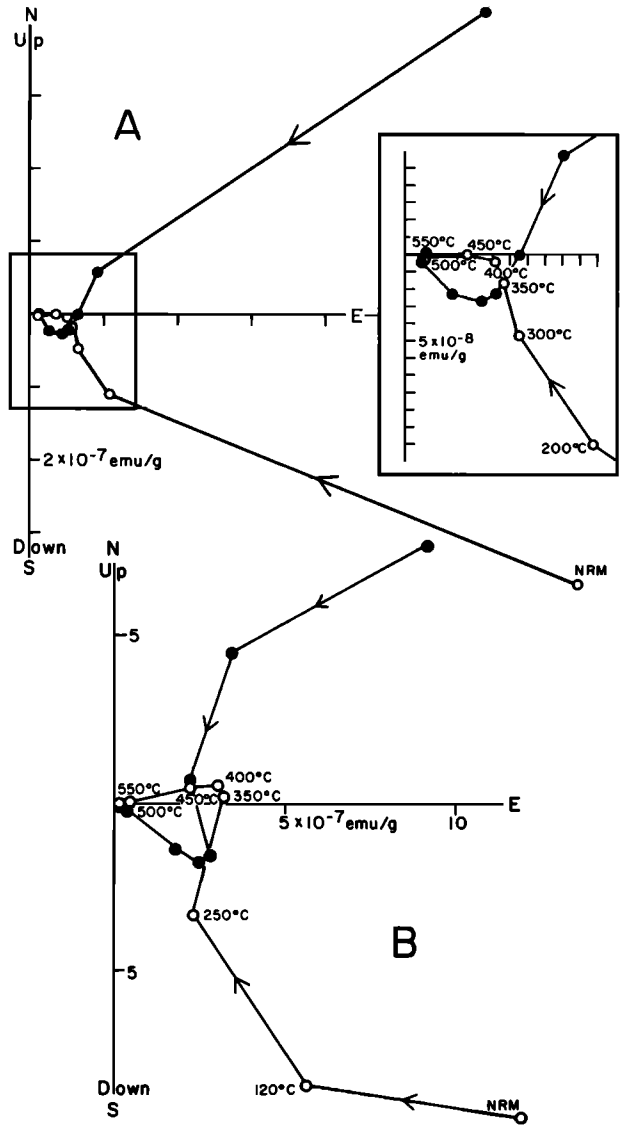


Fig. 6. Vector demagnetization diagrams of samples of gray, rusty-mottled limestone from the Carrara and lowermost Bonanza King Formations. The conventions used in constructing these diagrams are similar to those used by *Roy and Park* [1974]. Solid circles represent the endpoint of the magnetic vector projected onto the horizontal plane; the declination is the angle (measured clockwise from due north) between the north-south axis and the horizontal component. The distance from open circles to the origin is the total length of the magnetic vector; the angle between the abscissa and the ray from the origin through the open circle is the inclination. Note that open circles do not represent a projection of the magnetic vector onto a fixed plane. The demagnetization steps are labeled beside the points on the inclination path (1 emu/g = 1 A m²/kg). (*a*) Behavior upon thermal demagnetization of a sample from the Gold Ace Limestone Member of the Carrara Formation (group I). (*b*) Behavior upon thermal demagnetization of a sample from near the base of the Papoose Lake Member of the Bonanza King Formation (group II).

and II probably represents a vector sum of a component of intermediate thermal stability, the present axial field component, and the characteristic component. The exact direction of the component of intermediate thermal stability is difficult to determine because the relative intensities of all three components are not known. This component resembles a component with similar thermal stability found in upper Precambrian limestones at the base of the Desert Range section (D. R. Van Alstine and S. L. Gillett, unpublished data, 1977). The component with intermediate thermal stability is ascribed to the Sevier Orogeny of Late Cretaceous age.

Most important for interpreting the characteristic magnetization of both groups I and II is the identification of the magnetic species in which this component resides. Since the characteristic magnetization is removed above 500°C, it is possible that it is a DRM carried by magnetite. From the thermal demagnetization experiments alone, however, the possibility cannot be excluded that this component resides in hematite with a moderate blocking temperature. Accordingly, 25 duplicate specimens of gray, rusty-mottled limestones, all from group I, were subjected to progressive af demagnetization at peak fields up to 50 mT following thermal demagnetization at 410°C (above the thermal stability range of the two apparently secondary components but below that of the characteristic component). Vector demagnetization diagrams show that by 50 mT, af demagnetization has removed about 75% of the characteristic magnetization present after thermal demagnetization at 410°C (Figure 8). This suggests that the characteristic magnetization of these limestones resides in a species with a coercivity typical of magnetite and may well be a DRM. In addition, the specimen directions that were farthest from the present axial field direction at 410°C became better grouped upon af demagnetization (Figure 9).

Forty thin sections of gray, rusty-mottled limestone, mostly from the Red Pass and Pahrump Hills members of the Carrara Formation and including a few specimens which had been thermally demagnetized at 410°C, were examined in transmitted and reflected light. About 10 thin sections were also examined under cathodoluminescence. In addition, polished chips of about 90% of the samples from the Red Pass Member of the Carrara through the lowermost Papoose Lake Member of the Bonanza King Formation were examined under a binocular microscope.

Most of these rocks proved to be typical of the lime mudstone lithofacies of *Palmer and Halley* [1979], which makes up the greater part of the limestone members of the Carrara Formation. Fenestral fabric boundstones (characteristic of Palmer and Halley's algal boundstone lithofacies) and oolites are less common rock types; one intraclast conglomerate was also noted. No systematic correspondence was found between magnetic behavior and the specific limestone type. Staining of all rock types with slightly acidified solutions of alizarin red S (for calcite) and potassium ferricyanide (for Fe) [e.g., *Wolf et al.*, 1967] shows the rocks to be composed predominantly of nonferroan calcite. No fabric destruction was apparent in the samples that had undergone thermal demagnetization to 410°C; the only observed effect was to redden the orange seams discussed below, presumably from the dehydration of goethite to form hematite.

The rusty seams and mottling which lend these rocks their distinctive appearance in hand specimens are composed of cryptocrystalline, orange to red-orange iron oxides and oxyhydroxides ('limonite') which are probably dominantly goethite. The limonite seams contain occasional red streaks of hematite and abundant rhombs of dolomite. In some specimens, peloids or ooids are also commonly replaced by limonite. *Palmer and Halley* [1979] presented evidence suggesting that the oxidation of these rocks occurred late in their history. Hence the limonitic veins may well contain the magnetic species carrying the components of both low and intermediate thermal stability.

Examination of lightly etched surfaces on the thin sections under reflected light showed common grains of insoluble min-

21562202b, 1979, B9, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/JB084iB09p04475 by Curtin University Library, Wiley Online Library on [05/05/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

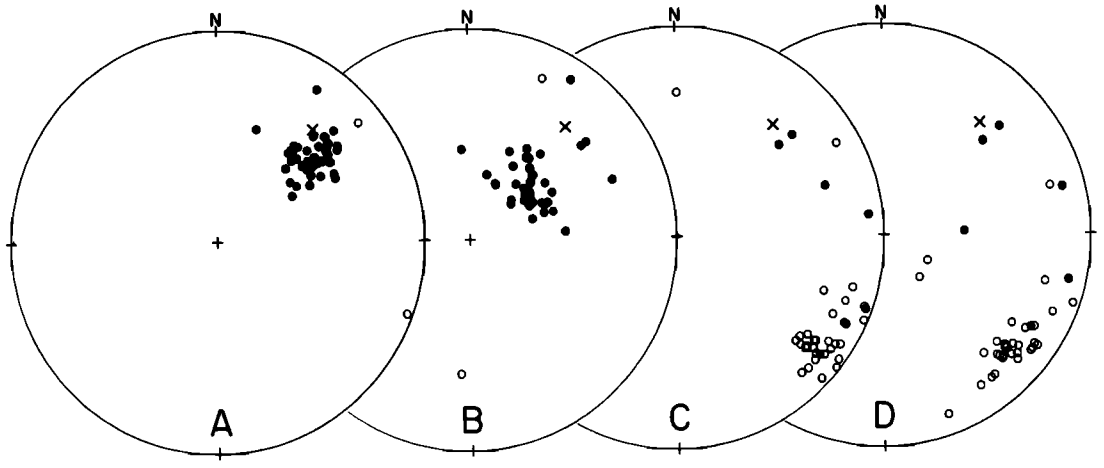


Fig. 7. Progressive thermal demagnetization behavior of all samples from group II (Jangle and Desert members of the Carrara Formation and lowermost Papoose Lake Member of the Bonanza King Formation). Symbols are as in Figure 3. (*a*) NRM, (*b*) 200° or 250°C, (*c*) 400°C, and (*d*) 450°C.

erals standing up in relief. Quartz and feldspar are most abundant, but some hematitic patches also occur. The euhedral form of most of the feldspars suggests they are authigenic. Under cathodoluminescence, however, many proved to have rounded, blue-luminescing cores. Such blue luminescence is due to a small amount of Ti^{4+} in the lattice [*Mariano et al.*, 1973] and is diagnostic of a high-temperature feldspar [*Kastner*, 1971]. Thus some of the feldspar grains contain detrital cores. The occurrence of such conclusively detrital material in these limestones makes it likely that a small component of detrital magnetite could also be present.

The characteristic magnetization direction for the group I samples (Table 1) was determined from specimens that had been subjected to af-following-thermal demagnetization (Figure 9). These demagnetization experiments yielded distributions of directions that are less biased toward the present axial field direction than the distributions obtained from thermal demagnetization alone. The characteristic magnetization direction represents the mode of combined specimen directions from the 20- and 35-mT steps. This procedure, which gives greater weight to those samples with more stable directions, was followed because the directions are about equally well grouped at both demagnetization steps. Axial symmetry of the highest probability contours around the mode and the consistency of this direction at two demagnetization steps suggest that the direction of characteristic magnetization has been isolated.

The characteristic magnetization direction of the group II samples (Table 1) was determined from the mode of directions combined from the 400° and 450°C demagnetization steps. Like the procedure employed above, this gives greater weight to those samples with more stable directions; it was adopted because the distributions of directions from both steps are about equally well grouped. The characteristic magnetization direction of the group II samples is within 5° of that for group I.

Secondary components of magnetization were completely removed from relatively few of the group I samples. However, analysis of vector demagnetization diagrams of all 185 samples permitted polarity assignments for about 60% of them. All samples that exhibited reasonably smooth thermal demagnetization paths are of reversed polarity; in addition, all group II samples not showing only a present field direction are unambiguously reversed. Hence the characteristic magnetization of the entire Carrara Formation and the lowermost Bonanza King may be entirely reversed (Figure 2). As the terrigenous units in the lower part of the group I strata (Eagle Mountain and Echo Shale members of the Carrara Formation and Resting Springs Member of the Zabriskie Quartzite) exhibit mostly scattered or present axial field directions, however, their polarity has been shown as 'not determinable' in Figure 2; intervals of normal polarity up to ~10 m thick might have been missed in these units.

The Bonanza King limestones from the stratigraphically higher section P not only are lithologically different from those at section O but also have very different magnetic characteristics. Section P contains dark gray to blue-gray limestone with common sparry calcite veins and seams. The 14 samples collected from this section had scattered NRM directions. Progressive af demagnetization to peak fields of 50 mT in general caused the directions to cluster more closely around the present axial field direction, with losses of magnetic intensity ranging from 50 to 90%. Moreover, progressive thermal demagnetization of five duplicate specimens yielded scattered directions. Hence no paleomagnetic information was obtained concerning any characteristic magnetization that might be present in this part of the Papoose Lake Member.

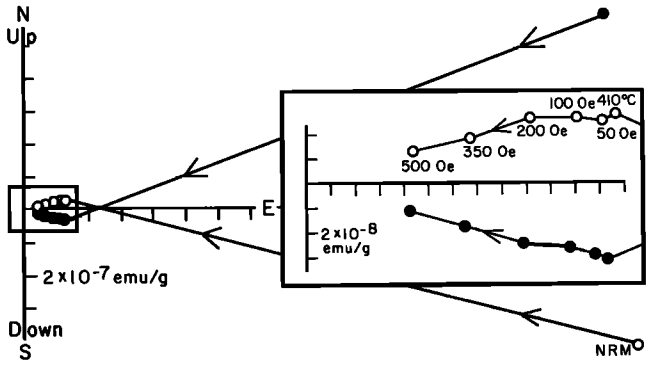


Fig. 8. Vector demagnetization diagram showing the behavior upon af-following-thermal demagnetization of a sample of gray, rusty-mottled limestone from the Pahrump Hills Member of the Carrara Formation (group I). Conventions are as in Figure 6 (1 Oe = 0.1 mT; 1 emu/g = 1 A m^2/kg).

21562202b, 1979, B9, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/JB084iB09p04475 by Curtin University Library, Wiley Online Library on [05/05/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

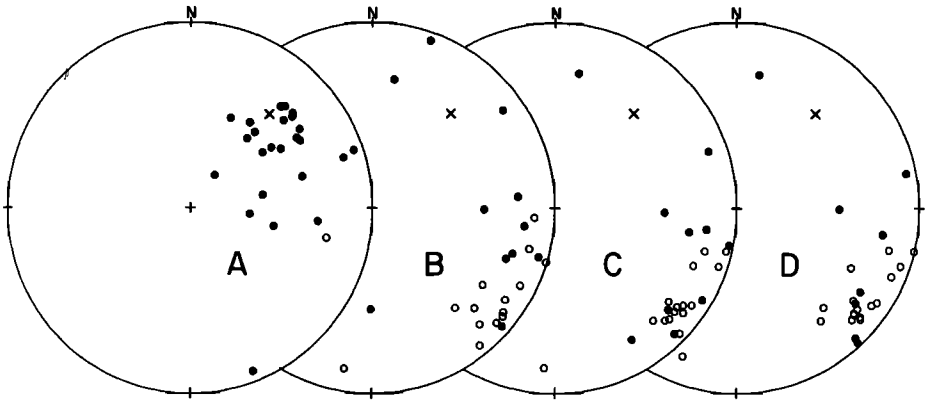


Fig. 9. Directions from combined af-following-thermal demagnetization experiments on duplicate specimens of gray, rusty-mottled limestone from group I (Carrara Formation up through Pahrump Hills Shale Member). Symbols are as in Figure 3. (*a*) NRM, (*b*) 410°C, (*c*) 410°C + 20 mT, and (*d*) 410°C + 35 mT.

DISCUSSION

The evidence presented above suggests that the characteristic magnetizations of the Desert Range Cambrian rocks were acquired penecontemporaneously with deposition. Additional support for penecontemporaneous acquisition is provided by the similarity of characteristic magnetization directions recorded by two apparently different magnetic species (in the Wood Canyon red-purple mudstones versus the gray, rusty-mottled limestones of the Carrara and Bonanza King). Because the Desert Range lies in a structurally complex region, however, caution must be exercised in interpreting paleomagnetic poles computed from these characteristic magnetizations. Since the rocks of the Desert Range may have been rotated about a vertical axis, poles derived from these rocks will not necessarily coincide with poles from rocks of equivalent age on the craton. In principle, the amount of vertical-axis rotation of the Desert Range can be determined by calculating the declination correction required to make paleomagnetic poles from these rocks match poles of the same age from the craton. The success of this technique depends upon the availability of accurate, well-dated Early to Middle Cambrian poles from the craton.

Most of the published paleomagnetic poles from North America that may represent Early and Middle Cambrian times are either insufficiently accurate or too imprecisely dated to be useful for comparison with the Desert Range Cambrian results (Table 2 and Figure 10). Poles 1 and 2, from extrusive and hypabyssal rocks of the Franklin Magnetic Interval in the Canadian Arctic, have a reported age of 625 m.y. B.P. from the least altered lavas [*Palmer and Hayatsu,* 1975]. Earlier workers [*Fahrig et al.*, 1971], however, preferred an age of 675 m.y. B.P. for these rocks. Even the younger age seems somewhat too old for middle Wood Canyon time, as the base of the Cambrian is probably in the range 590 ± 20 m.y. B.P. [*Lambert,* 1971]. Pole 3 from three flows of the Cloud Mountain

TABLE 2. Possibly Early and Middle Cambrian Paleomagnetic Poles From North America

Pole	Pole Position	Rock Unit and Location	Age	Reference
14	37°N, 122°E	Intrusives, Colorado	Late Precambrian to Early Ordovician (704–485 m.y. B.P.)	*Larson and Mutschler* [1971], *French et al.* [1977]
13	44°N, 100°E			
12	15°N, 142°E			
11	5°N, 174°E			
10	48°N, 107°E			
9	59°N, 89°E	Abrigo Formation, Arizona	Middle to Late Cambrian	*Elston and Bressler* [1977]
8	55°N, 110°E	Muav Formation, Arizona	Middle to Late Cambrian	*Elston and Bressler* [1977]
7	38°N, 144°E	Rome Formation, Tennessee	Middle Cambrian	*French* [1976]
6	13°N, 146°E	Ophiolite complex, Quebec	Middle(?) Cambrian (550(?) m.y. B.P.)	*Seguin* [1976]
5	5°N, 158°E	Tapeats Sandstone, Arizona	Early to Middle Cambrian	*Elston and Bressler* [1977]
4	29°N, 167°E	Bradore Formation, western Newfoundland	Early Cambrian	*Rao and Deutsch* [1976]
3	5°N, 172°E	Cloud Mountain Basalt, western Newfoundland	Late Precambrian to Early Cambrian (605 ± 10 m.y. B.P.)	*Deutsch and Rao* [1977]
2	4°S, 161°E	Franklin rocks, Canadian Arctic (Group A)	Late Precambrian (675–625 m.y. B.P.)	*Palmer and Hayatsu* [1975]
1	8°N, 166°E	Franklin rocks, Canadian Arctic (Group B)		

21562202b, 1979, B9, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/JB084iB09p04475 by Curtin University Library, Wiley Online Library on [05/05/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

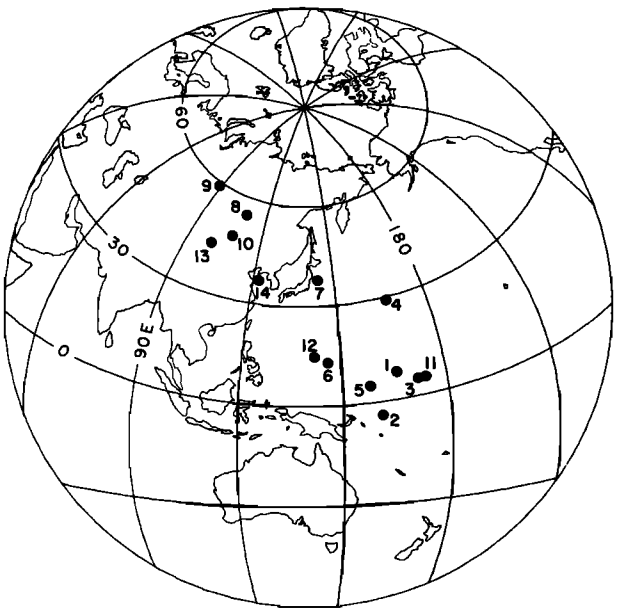


Fig. 10. Published, possibly Early and Middle Cambrian paleomagnetic poles from North America. Pole numbers are as in Table 2.

Basalt, western Newfoundland, may also be older (605 ± 10 m.y.) than middle Wood Canyon time and may have incompletely averaged out secular variation [*Deutsch and Rao,* 1977]. In addition, the possibility of a significant rotation of western Newfoundland with respect to cratonic North America has been the subject of a long-standing controversy [cf. *Deutsch and Rao,* 1977]. This uncertainty also applies to pole 4, from the Early Cambrian Bradore Formation of western Newfoundland. Moreover, the attitude correction upon which the Bradore pole is based is ambiguous [*Rao and Deutsch,* 1976]. Pole 6, from a Middle(?) Cambrian ophiolite complex in Quebec [*Seguin,* 1976], is derived from rocks in a structurally complex area and may reflect a significant vertical-axis rotation. Pole 7, from the Rome Formation, Tennessee, is from a thrust sheet, which may also have rotated about a vertical axis [*French,* 1976]. Although poles 8 and 9 from the Muav and Abrigo formations, Arizona, are from rocks in structurally simpler settings, on faunal evidence these strata are younger (latest Middle and earliest Late Cambrian) than early Bonanaza King time and, in addition, may have been remagnetized [*Elston and Bressler,* 1977]. Intrusives from Colorado studied by *Larson and Mutschler* [1971] and *French et al.* [1977] yield ages ranging from 704 to 485 m.y. B.P. and give a multimodal distribution of paleomagnetic directions (poles 10–14).

Of the poles listed in Table 2, the pole from the Tapeats Sandstone of the Colorado Plateau (pole 5) is the most relevant for comparison with the Desert Range Cambrian paleomagnetic results. The Tapeats pole, which has an α_{95} of 3.3°, is based on 129 samples representing antipolar normal and reversed directions. The Tapeats is unfossiliferous except locally at its top in fine-grained beds gradational in lithology with the overlying Bright Angel Shale. In the westernmost exposures of a typically developed Tapeats section at Frenchman Mountain, Nevada, such transitional beds yield trilobites of latest Early Cambrian age, equivalent to those in the lower Pyramid Shale Member of the Carrara (A. R. Palmer, written communication, 1976). The Tapeats, however, becomes younger west to east so that its upper part is probably early Middle Cambrian in the eastern Grand Canyon [*McKee and Resser,* 1945], where the pole of *Elston and Bressler* [1977] was derived. This implies that the section represented by the Tapeats pole should be partly older than and partly contemporaneous with the section represented by the group I pole from the Carrara Formation, since the group I pole was derived from rocks that are mainly early Middle Cambrian in age. Moreover, on the basis of lithologic comparison and regional stratigraphic relationships, *Stewart* [1970] has suggested that the Tapeats is the rock-stratigraphic equivalent of the Zabriskie Quartzite and upper part of the Wood Canyon Formation; therefore the entire Tapeats almost certainly is younger than the middle member of the Wood Canyon. In addition, preliminary paleomagnetic results from the Bright Angel Shale, which conformably overlies the Tapeats, suggest that a pole from the Bright Angel would be similar to that from the Tapeats [*Elston and Bressler,* 1977]. The Bright Angel extends into the Middle Cambrian *Glossopleura* Zone at the top [*McKee and Resser,* 1945] and therefore is partly contemporaneous with the upper Carrara and lowermost Bonanza King Formations. In summary, lithostratigraphic and biostratigraphic evidence strongly indicate that the age of the Tapeats is bracketed by the Desert Range strata.

However, poles computed from the characteristic magnetization directions of the Early and Middle Cambrian strata from the Desert Range differ by about 35° from the Tapeats pole (Figure 11). The direction of characteristic magnetization for the Tapeats (D = 94°, I = −6°) differs almost entirely in declination from the directions for the middle Wood Canyon (D = 124°, I = −3°) and the Carrara group I samples (D = 130°, I = −14°). Hence the discordance between Cambrian poles from the Desert Range and the pole from the Tapeats can be explained by clockwise rotation of the entire Desert Range Cambrian section. A 36° counterclockwise rotation of the Desert Range declinations will place the pole from the Carrara group I samples (7°N, 158°E) and that from the Wood Canyon (1°S, 157°E) on either side of the pole from the Tapeats (5°N, 158°E). A vertical-axis rotation of this magnitude is entirely consistent with the structural evidence (summarized in the discussion of the regional geology) for clockwise oroflexural bending during the Cenozoic in the southern Great Basin. Because the Desert Range is part of a thrust plate, however, some component of the net 36° clockwise rotation may have resulted from thrusting in the Mesozoic. Significant vertical-axis rotations have been documented by paleomagnetic investigations in other thrust systems of the Cordillera and have amplitudes up to 60° in Wyoming [*Grubbs and Van der Voo,* 1976].

The polarity of the characteristic magnetizations of the Desert Range Cambrian rocks is consistent with the polarity zonation observed in Cambrian rocks of the Colorado Plateau. *Elston and Bressler* [1977] have determined that several zones of reversed and normal polarity are represented in the lower and middle parts of the Tapeats Sandstone, whereas the uppermost Tapeats and overlying Bright Angel Shale seem to be entirely reversely magnetized. As discussed previously, the Tapeats is possibly the time equivalent of the upper Wood Canyon through lowermost Carrara Formations. Perhaps the zones of normal polarity inferred in the Zabriskie correlate with some of those found in the Tapeats, and the long reversed interval in the Carrara and lowermost Bonanza King correlates with the reversed interval in the uppermost Tapeats and Bright Angel. The reversed interval in the middle member of the Wood Canyon occurs in rocks that are probably older than the basal Tapeats [*Stewart,* 1970].

21562202b, 1979, B9, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/JB084iB09p04475 by Curtin University Library, Wiley Online Library on [05/05/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License



Fig. 11. Cambrian paleomagnetic poles from the Desert Range before (unprimed symbols) and after (primed symbols) correction for the proposed 36° clockwise vertical-axis rotation of the Desert Range section with respect to the Colorado Plateau. II, group II (uppermost Carrara and lowermost Bonanza King Formations; I, group I (remainder of Carrara Formation); WC, Wood Canyon Formation. Star labeled 'T' is the pole from the Tapeats Sandstone of the Colorado Plateau [*Elston and Bressler,* 1977]. Cross labeled 'DR' marks the sampling site. Also shown is the post-Early-Ordovician part of the apparent polar wander path for North America determined by *Van Alstine* [1979]. Numbers beside the path refer to nearly uniform, 22-m.y. time intervals: 1–3, Cenozoic; 4–7, Cretaceous; 7–9; Jurassic; 10–11, Triassic; 12–13, Permian; 14–15, Pennsylvanian; 16–17, Mississippian; 18–19, Devonian; 20–21, Silurian; and 21–22, Ordovician.

The Cambrian polarity history beginning to emerge from paleomagnetic studies in the southwestern United States is supported by results from other regions. D. R. Van Alstine (unpublished data, 1977) has found only reversed polarity in rocks of the Canadian Cordillera spanning most of Middle Cambrian time. In addition, *Irving and Pullaiah*'s [1976] synthesis of worldwide polarity data suggests that the geomagnetic field during most of the Cambrian was characterized by a strong reversed-polarity bias.

CONCLUSIONS

Paleomagnetic evidence suggests that part of the Desert Range has undergone 36° of net clockwise rotation with respect to the Colorado Plateau. The existence of such large vertical-axis rotations implies that paleomagnetic results from tectonically complicated areas such as the southern Great Basin must be treated with discretion. However, such results also demonstrate the value of paleomagnetism in establishing constraints on structural geologic problems.

The similarity of Early and Middle Cambrian poles from the Desert Range strata strongly suggests that the paleogeographic pole was relatively stationary with respect to North America between middle Wood Canyon (early Early Cambrian) and early Bonanza King (middle Middle Cambrian) time.

A distinctive Cambrian polarity zonation seems to be emerging from paleomagnetic investigations of Early Paleozoic rocks. A change from mixed to dominantly reversed polarity appears to have occurred near the Early-Middle Cambrian boundary and may form a very useful worldwide chronologic horizon.

REFERENCES

Albers, J. P., Belt of sigmoidal bending and right lateral faulting in the western Great Basin, *Geol. Soc. Amer. Bull., 78,* 143–156, 1967.

Burchfiel, B. C., Structural geology of the Specter Range Quadrangle, Nevada, and its regional significance, *Geol. Soc. Amer. Bull., 76,* 175–192, 1965.

Burchfiel, B. C., P. J. Pelton, and J. Sutter, An early Mesozoic deformation belt in south-central Nevada–southeastern California, *Geol. Soc. Amer. Bull., 81,* 211–215, 1970.

Burchfiel, B. C., R. J. Fleck, D. T. Secor, R. R. Vincelette, and G. A. Davis, Geology of the Spring Mountains, Nevada, *Geol. Soc. Amer. Bull., 85,* 1013–1022, 1974.

Deutsch, E. R., and K. V. Rao, New palaeomagnetic evidence fails to support rotation of western Newfoundland, *Nature, 266,* 314–318, 1977.

Ekren, E. B., C. L. Rogers, R. E. Anderson, and P. P. Orkild, Age of Basin and Range normal faults in Nevada Test Site and Nellis Air

21562202b, 1979, B9, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/JB084iB09p04475 by Curtin University Library, Wiley Online Library on [05/05/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

Force Range, Nevada, Nevada Test Site, *Geol. Soc. Amer. Mem.*, 110, 1968.
Elston, D. P., and S. L. Bressler, Paleomagnetic poles and polarity zonation from Cambrian and Devonian strata of Arizona, *Earth Planet. Sci. Lett., 36,* 423–433, 1977.
Fahrig, W. F., E. Irving, and G. D. Jackson, Paleomagnetism of the Franklin Diabases, *Can. J. Earth Sci., 8,* 455–467, 1971.
Fisher, R. A., Dispersion on a sphere, *Proc. Roy. Soc., Ser. A, 217,* 295–305, 1953.
Fleck, R. J., Tectonic style, magnitude, and age of deformation in the Sevier orogenic belt in southern Nevada and eastern California, *Geol. Soc. Amer. Bull., 81,* 1705–1720, 1970.
French, R. B., Lower Paleozoic paleomagnetism of the North American craton, Ph.D. thesis, Univ. of Mich, Ann Arbor, 1976.
French, R. B., D. H. Alexander, and R. Van der Voo, Paleomagnetism of upper Precambrian to lower Paleozoic intrusives from Colorado, *Geol. Soc. Amer. Bull., 88,* 1785–1792, 1977.
Grubbs, K. L., and R. Van der Voo, Structural deformation of the Idaho-Wyoming overthrust belt (U.S.A.), as determined by Triassic paleomagnetism, *Tectonophysics, 33,* 321–336, 1976.
Irving, E., and G. Pullaiah, Reversals of the geomagnetic field, magnetostratigraphy, and relative magnitude of paleosecular variation in the Phanerozoic, *Earth Sci. Rev., 12,* 35–64, 1976.
Kastner, M., Authigenic feldspars in carbonate rocks, *Amer. Mineral., 56,* 1403–1442, 1971.
Lambert, R., The pre-Pleistocene Phanerozoic time-scale—A review, The Phanerozoic Time-Scale—A Supplement, *Spec. Publ. 5,* The Geol. Soc., London, 1971.
Larson, E. E., and F. E. Mutschler, Anomalous paleomagnetic pole from isotopically dated Cambro-Ordovician intrusives in Colorado, *Geol. Soc. Amer. Bull., 82,* 1657–1666, 1971.
Longwell, C. R., The mechanics of orogeny, *Amer. J. Sci., 243-A,* 417–447, 1945.
Longwell, C. R., Possible explanation of diverse structural patterns in southern Nevada, *Amer. J. Sci., 258-A,* 192–203, 1960.
Longwell, C. R., Measure and date of movement on the Las Vegas Valley Shear Zone, Clark County, Nevada, *Geol. Soc. Amer. Bull., 85,* 985–990, 1974.
Longwell, C. R., E. H. Pampeyan, B. Bowyer, and R. J. Roberts, Geology and mineral deposits of Clark County, Nevada, *Nev. Bur. Mines Bull., 62,* 1965.
Mariano, A. N., J. Ito, and P. J. Ring, Cathodoluminescence of plagioclase feldspars, *Geol. Soc. Amer. Abstr. Programs, 5,* 726, 1973.
McElhinny, M. W., and B. J. J. Embleton, Precambrian and Early Palaeozoic palaeomagnetism in Australia, *Phil. Trans. Roy. Soc. London, Ser. A, 280,* 417–431, 1976.
McElhinny, M. W., J. W. Giddings, and B. J. J. Embleton, Palaeomagnetic results and late Precambrian glaciations, *Nature, 248,* 557–561, 1974.
McKee, E. D., and C. E. Resser, Cambrian history of the Grand Canyon region, *Carnegie Inst. Wash. Publ., 563,* 1945.
Nelson, C. A., Late Precambrian–Early Cambrian stratigraphic and faunal succession of eastern California and the Precambrian-Cambrian boundary, *Geol. Mag., 115,* 121–126, 1978.
Palmer, A. R., The Cambrian of the Great Basin and adjacent areas, western United States, in *Cambrian of the New World,* edited by C. H. Holland, pp. 1–78, John Wiley, New York, 1971.
Palmer, A. R., and R. B. Halley, Physical stratigraphy and trilobite biostratigraphy of the Carrara Formation (Lower and Middle Cambrian) in the southern Great Basin, *U.S. Geol. Surv. Prof. Pap., 1047,* in press, 1979.
Palmer, H. C., and A. Hayatsu, Paleomagnetism and K-Ar dating of some Franklin lavas and diabases, Victoria Island, *Can. J. Earth Sci., 12,* 1439–1447, 1975.
Rao, K. V., and E. R. Deutsch, Paleomagnetism of the Lower Cambrian Bradore Sandstones, and the rotation of Newfoundland, *Tectonophysics, 33,* 337–357, 1976.
Roy, J. L., and J. K. Park, The magnetization process of certain red beds: Vector analysis of chemical and thermal results, *Can. J. Earth Sci., 11,* 437–471, 1974.
Seguin, M. K., Reconnaissance paleomagnetic investigation of the ophiolitic complex, Thetford Mines, Asbestos, Quebec, *Tectonophysics, 34,* 231–243, 1976.
Stewart, J. H., Possible large right-lateral displacement along fault and shear zones in the Death Valley–Las Vegas area, California and Nevada, *Geol. Soc. Amer. Bull., 78,* 131–142, 1967.
Stewart, J. H., Upper Precambrian and Lower Cambrian strata in the southern Great Basin, California and Nevada, *U.S. Geol. Surv. Prof. Pap., 620,* 1970.
Stewart, J. H., and H. Barnes, Precambrian and Cambrian formations in the Desert Range, Clark County, Nevada, *U.S. Geol. Surv. Bull., 1244-A,* A35–A42, 1966.
Strangway, D. W., *History of the Earth's Magnetic Field,* 168 pp., McGraw-Hill, New York, 1970.
Van Alstine, D. R., Apparent polar wandering with respect to North America since the late Precambrian, Ph.D. thesis, Calif. Inst. of Technol., Pasadena, 1979.
Van Alstine, D. R., and J. de Boer, A new technique for constructing apparent polar wander paths and the revised Phanerozoic path for North America, *Geology, 6,* 137–139, 1978.
Van Alstine, D. R., and S. L. Gillett, Paleomagnetism of upper Precambrian sedimentary rocks from the Desert Range, Nevada, *J. Geophys. Res., 84,* this issue, 1979.
Winograd, I. J., and W. Thordarson, Structural control of groundwater movement in miogeosynclinal rocks of south-central Nevada, Nevada Test Site, *Geol. Soc. Amer. Mem., 110,* 1968.
Wolf, K. H., A. J. Easton, and S. Warne, Techniques of examining and analyzing carbonate skeletons, minerals, and rocks, in *Carbonate Rocks,* edited by G. V. Chilingar, H. J. Bissel, and R. W. Fairbridge, pp. 253–341, Elsevier, New York, 1967.
Zijderveld, J. D. A., A.c. demagnetization of rocks: Analysis of results, in *Methods in Palaeomagnetism,* edited by D. W. Collinson, K. M. Creer, and S. K. Runcorn, pp. 254–286, Elsevier, New York, 1967.

(Received November 13, 1978;
revised March 8, 1979;
accepted March 12, 1979.)

21562202b, 1979, B9, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/JB084iB09p04475 by Curtin University Library, Wiley Online Library on [05/05/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License